Algebra II with Trigonometry Chapter 5.1

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4401267_alg-2trig-h-chp-5-1-note-docx.pdf

Problems

1. Find the smallest positive degree measure that is coterminal with an angle of exactly $1,000,000^\circ$.

2. A wire of fixed length L is bent to form the complete boundary (two radii and an arc) of a circular sector. What central angle (in radians) maximizes the enclosed area of this sector?

3. Circle A has a radius of 1 unit and Circle B has a radius of 2024 units. If Circle A rolls without slipping around the stationary, outer circumference of Circle B, how many full 360° revolutions will Circle A have completed about its *own* center by the time it returns to its exact starting position?

4. A single central angle θ is drawn within a pair of concentric circles, dividing the shape into a smaller inner circular sector and a surrounding outer annular region. If these two regions have the exact same area *and* the exact same perimeter, what is the value of θ in radians?

5. Assuming the Earth is a perfect sphere of radius R, a city on the equator travels through space at a linear speed v purely due to the Earth's rotation. A second city, located perfectly due north of the first, travels at a linear speed of exactly $v/2$. What is the shortest distance along the Earth's surface between the two cities?

6. At 12:00, the minute and hour hands of a continuously moving clock perfectly overlap. Over the exact span of the next 12 hours, how many times will the angle between the two hands measure exactly 1 radian?